

Proposal

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Contributing authors: ;

1 Introduction

Type 1 diabetes (T1D) is an autoimmune condition resulting in absolute insulin deficiency and a lifelong requirement for insulin replacement [1]. Glycemic control in T1D remains challenging, with only approximately 20% of American adults and 30% of Europeans achieving an HbA1c target of $< 7\%$ [2], despite advancements in insulin analogs [3], enhanced accuracy of continuous glucose monitoring (CGM) [4], increased adoption of intensive insulin therapy [5], and the emergence of hybrid automated insulin delivery (hAID) systems. While these technologies have demonstrated benefits in mitigating diabetes-related complications [6] and potentially reducing excess mortality in certain populations [7], mortality and complication rates in T1D still exceed those in the general population [8].

We conduct a comprehensive evaluation on benchmark using RMSE for regression,. Our preliminary results reveal that, based on RMSE metrics, NHITS, PatchTST, and TimeLLM do not demonstrate any improvement over the established benchmark, Long Short-Term Memory (LSTM), with the advanced models yielding higher average RMSE values (e.g., 37.73 mg/dL for NHITS, 38.64 mg/dL for PatchTST, and 40.96 mg/dL for TimeLLM compared to 36.72 mg/dL for LSTM). This lack of improvement is attributed to temporal shifts in predictions, where the advanced models capture intricate patterns and subtle fluctuations in blood glucose dynamics more faithfully, yet incur penalized RMSE scores due to offsets from actual timings. Despite these initial metrics, our innovative approach highlights potential for refinements to achieve substantial advancements in prediction accuracy, establishing a new foundation for blood glucose forecasting with significant implications for real-world clinical applications.

2 Datasets and task

In this section, we first outline the key characteristics of the two datasets employed in this study. Subsequently, we formally define the research task within a time series framework and specify the selected evaluation metrics.

2.1 Datasets

In this paper, we utilize continuous glucose monitoring (CGM) data from two distinct cohorts of individuals with type 1 diabetes (T1D), each gathered under varying study designs, population sizes, and data collection conditions.

The first dataset, T1DATA, stems from a randomized crossover clinical trial involving 35 individuals with T1D [9]. Participants used Dexcom G6 CGM devices, recording glucose levels at five-minute intervals over approximately 30 days. The cohort comprised 23 women and 12 men, with a mean age of 44.5 ± 15.4 years and an average HbA1c of $6.7 \pm 0.9\%$.

The second dataset, T1DEXI, derives from the Type 1 Diabetes Exercise Initiative—a large-scale observational study involving 497 individuals with T1D [10]. CGM data were collected using either personal or blinded Dexcom G6 devices at five-minute intervals over approximately 28 days, adhering to a protocol similar to T1DATA. The cohort had a mean age of 36.7 ± 14 years and an average HbA1c of $6.6 \pm 0.7\%$, consisting of 383 women and 134 men.

2.2 Prediction tasks

Time series prediction [11] entails using historical data to forecast future outcomes of a phenomenon. In this study, we regard patient data from CGM devices, which monitor blood glucose levels over time, as the phenomenon to predict, employing a rolling prediction strategy. We denote the number t of future samples to predict as the *Horizon Window* (HW). Given that samples occur every 5 minutes in both datasets, the horizon window duration equates to $5t$ minutes. In this setup, two scenarios are feasible: a regression task and a classification task.

Predicting the t actual blood glucose levels within the horizon window constitutes a classic regression problem. Model performance is assessed using the *root mean square error* (RMSE), a standard metric for evaluating regression systems, particularly in blood glucose forecasting [12]. RMSE quantifies the difference between ground truth and predicted values, defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad , \quad (1)$$

where n is the total number of values, and y_i and $\hat{y}_i$ denote the ground truth and predicted values at time step i , respectively.

3 Pre-processing and models

This section first details the architecture of the baseline model and the advanced models with diverse core architectures for blood glucose level predictions. Second, we describe the pre-processing steps applied to both datasets (T1DATA and T1DEXI) and the training configurations.

3.1 Models

As the baseline, we employ the Long Short-Term Memory (LSTM) network [13], a recurrent neural network (RNN) designed for time-dependent data. The LSTM features a chain-like structure of repeating modules, or cell states, through which information flows. It incorporates a gating mechanism to regulate information addition or removal from the cell state, comprising three primary gates: (1) the forget gate, which determines retained information via a sigmoid layer; (2) the input gate, which filters incoming information; and (3) the output gate, which controls outgoing information.

LSTMs are established in blood glucose prediction [12, 14, 15]. However, they struggle to capture specific features of blood glucose signals over extended prediction horizons. To address this limitation, we propose employing three advanced forecasting models with distinct core architectures: NHITS (MLP-based), PatchTST (Transformer-based), and TimeLLM (LLM-based), all suited for longer horizons.

- **NHITS** [16] is an MLP-based model extending NBEATS [17]. It employs multi-resolution decomposition, breaking the input signal into various frequencies to capture short-term variations and long-term trends. The architecture consists of stacked fully connected layers divided into blocks, each learning a specific frequency. Through progressive refinement, each block enhances predictions by adjusting prior residuals. NHITS aggregates multi-resolution components for the final prediction.
- **PatchTST** [18] is a Transformer-based model for long-term time series forecasting. It decomposes multivariate inputs into univariate series. Unlike traditional Transformers with point-wise attention, PatchTST uses a patching mechanism, segmenting the series into patches. This enables group-level analysis, enhancing local pattern detection and reducing computational load. A standard Transformer encoder processes patches, extracting multi-scale representations. A linear layer generates predictions, integrated with contextual information for improved accuracy.
- **TimeLLM** [19] leverages large language models (LLMs) [20] for time series forecasting. Rather than fine-tuning LLM weights, it introduces a patch embedding layer to tokenize raw series into interpretable patches. A reprogramming layer reframes the task as language-like, enabling LLM processing. Optional prompts enhance reasoning. A projection layer converts LLM outputs to numerical predictions, facilitating efficient long-term forecasting.

Table 1 presents the average number of parameters and model size for each architecture post-hyperparameter optimization (trained on T1DATA).

LSTM and NHITS exhibit similar sizes and parameter counts, while PatchTST has the fewest parameters. TimeLLM is two to three orders of magnitude larger, posing challenges for real-time deployment on wearable devices.

3.2 Pre-processing and training configuration

Training and evaluation for both datasets use a fixed horizon window of $t = 12$ samples (60 minutes). We rely solely on blood glucose signals for predictions. Missing values are imputed via linear interpolation in both datasets.

Table 1 Comparison of parameters and model size for LSTM, NHITS, PatchTST, and TimeLLM.

Model	Parameters ($\times 10^6$)		Size (MB)
	LSTM	2.7	10.33
	NHITS	2.4	9.26
	PatchTST	0.4	1.54
	TimeLLM	216.9	827.41

Models are trained using mean squared error (MSE) as the loss function (MSE computes average squared differences; RMSE, its square root, aids interpretability in original units).

For T1DATA, patient-specific models are trained with individualized hyperparameter optimization. Data are split into training (60%), validation (20%), and testing (20%) subsets, preserving temporal order to prevent leakage.

4 Empirical evaluation

4.1 Regression results

For regression, we start with T1DATA using patient-specific models, then examine transfer learning on both datasets.

4.1.1 Individual models per patient

Table 2 shows average RMSE (mg/dL) across T1DATA patients for each model, including standard deviation.

TimeLLM yields the highest average RMSE at 40.96 mg/dL. PatchTST and NHITS achieve 37.73 mg/dL and 38.64 mg/dL, respectively. The LSTM baseline records the lowest at 36.72 mg/dL, initially suggesting superior performance.

However, these findings require qualification. As illustrated in Fig. 1, a prediction shift emerges, where LSTM exhibits less fidelity to signal granularity but benefits from the shift in RMSE calculation. In contrast, advanced models track the signal more closely, yet incur higher RMSE due to temporal offsets.

Table 2 Average RMSE (mg/dL) over T1DATA patients for LSTM, NHITS, PatchTST, and TimeLLM.

Model	RMSE (std)
LSTM	36.72 $\pm$ 8.21
NHITS	37.73 $\pm$ 7.82
PatchTST	38.64 $\pm$ 8.10
TimeLLM	40.96 $\pm$ 9.28

Time Series Plot for 85101 - HW: 12

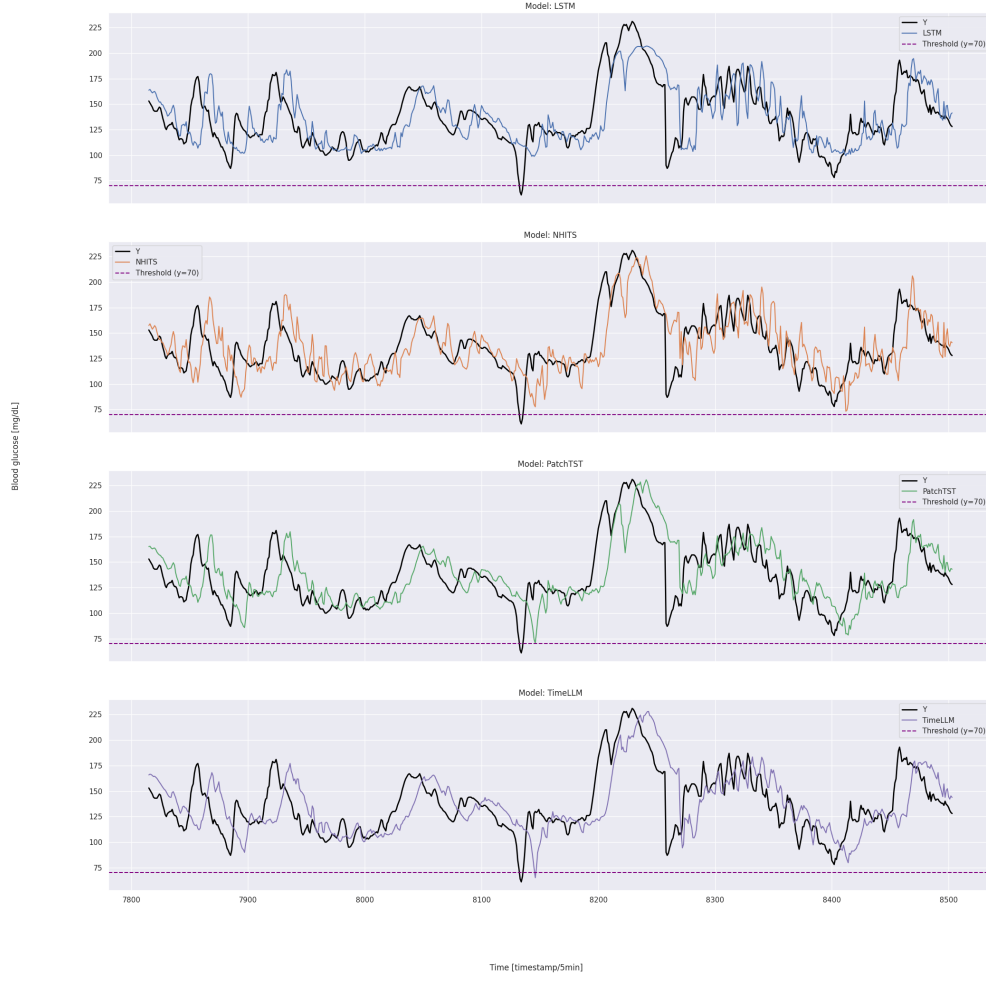


Fig. 1 Predictions across all four models.

5 Proposed research

5.1 New Metric

Beyond regression, detecting specific events within the horizon window, particularly hypoglycemic events (blood glucose below 70 mg/dL), is of interest. This involves predicting the occurrence of such an event in the next $5t$ minutes—a binary classification task. To evaluate using metrics like precision or recall, we identify two key time points.

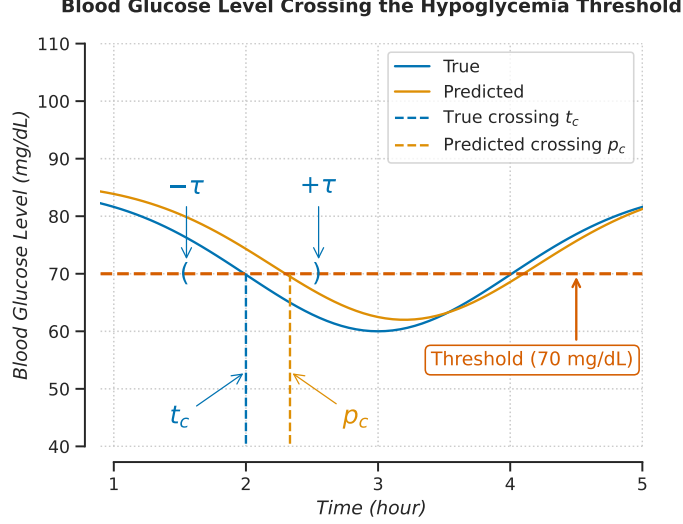


Fig. 2 Hypoglycemia detection: True (blue) and predicted (orange) blood glucose levels, with true t_c (blue dashed) and predicted p_c (orange dashed) crossings of the 70 mg/dL threshold (red dashed). The τ parameter relaxes the crossing constraint.

The true crossing time t_c and predicted p_c are shown in Fig. 2. A parameter τ relaxes the match: if $p_c \in [t_c - \tau, t_c + \tau]$, it is a True Positive (TP). False Positives (FP) occur when no true event exists but one is predicted in the interval; False Negatives (FN) when a true event is missed.

Performance metrics are:

- Precision (Pr): $\frac{TP}{TP+FP}$
- Recall (Re): $\frac{TP}{TP+FN}$
- F1-score (F1): $2 \cdot \frac{Pr \cdot Re}{Pr+Re}$

5.2 Reducing prediction shift

Preliminary experiments indicate systematic deviations in predictions over extended horizons. Advanced models like PatchTST, NHITS, and TimeLLM capture blood glucose fluctuations more accurately than LSTM but exhibit temporal offsets, inflating RMSE despite proximity to actual values.

Three promising avenues to address this:

- Offset-aware loss functions: Integrate metrics like cross-correlation or Dynamic Time Warping (DTW) into losses to balance accuracy and temporal alignment.
- Multi-step prediction: Train to forecast intermediate values (e.g., y_{t+1} to y_{t+12}), enhancing trend capture and reducing offsets.
- Extended temporal context: Input longer sequences (e.g., 2–3 hours) for richer context, improving future dynamics prediction.

These strategies outline a robust research agenda to enhance temporal consistency, minimize offsets, and maximize blood glucose prediction efficacy.

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